Checkpoint E - Delivery, Observation, Recovery

QuantRule

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Claim

Claim type: Outcome.

> For one researcher, a submitted run completes without the client and is reproducible from its own provenance, measured by comparing the canonical digest of a rerun on a second day, with the limitation that this says nothing about the rule's edge.

Delivery

- One image, six binaries, one digest.
- One command: `make up` -> healthy -> a URL.
- One published port.
- A desktop shell installs from a single release asset and reads no local data.

Observation

Signal	Where	Action when it moves
Service state	`/health` per dependency	restart the failing dependency
Queue depth, workers	`/health`	add a worker
Run state and failed stage	run detail	fix the input, resubmit
Canonical digest	run detail, `make verify`	compare; a mismatch is a defect
Image digest	`/health`, provenance	a different build is a different run

The run row is the record. A dead-lettered message is visible as a failed run, not as a silent loss.

Recovery

Failure	Recovery	Evidence it worked
Worker killed mid-run	message redelivered; recompute is idempotent	no artifact visible; rerun yields the same digest
Evidence service down after compute	finalization retries	run stays `running`, never `succeeded` with a missing artifact
Database lost	restore the last backup	every artifact hash verifies

Failure	Recovery	Evidence it worked
Object store lost	restore from backup	missing objects make runs unverifiable, and that is reported
Bad image	start the previous digest	migrations are forward-only; the previous version must tolerate the newer schema

Limit

There is no high availability and no failover. A host failure is an outage until the operator restarts the stack. That is a deliberate trade for a single-operator tool, and it is recorded, not hidden.

Decision

Ship the thin shell and profiles with v1.0.0; keep the self-contained desktop as v1.1 behind the adapter traits. Next action: prove the delivered run survives a client close and a worker kill, then verify the digest on a second machine.